Wilderfolk v0.6.5.1 — Complete Unabridged Changelog

Official Release Ledger & Feature Index (Released October 19, 2026) — 100% Complete Record

Total Logged Updates: *200+ Entries*	***Major Additions:*** *Villa, Square, Graveyard, Gazette*	***Map Scaling:*** *Up to 6144×4608 Huge Maps*

New Buildings, Housing & Architecture

- **Fixed:** synchronized the asset register with the shipped Village Square artwork (square.jpg) and updated the Square's sprite path; Graveyard and Villa are also marked as used by the building renderer.
- **Added:** automated Villa housing regression tests covering Better Housing unlocks, build-catalog registration, 14/16/18 resident capacities, the level-3 upgrade ceiling, and unchanged House/Mansion capacity rules.
- **Added:** Villa residence using /sprites/villa.png with shelter for 14 villagers; listed under Housing and included in residential occupancy and population housing capacity. Building a Villa requires the new **Better Housing** research in the Architecture tree (after Fine Construction), including on older saves. Villa upgrades follow the standard housing rules: level 1 shelters 14 people, level 2 shelters 16 and level 3 shelters 18.
- **Added:** a unique Village Square using /sprites/square.jpg, serving as the gathering place for festivals and elections, with the previous gathering locations retained when no completed square exists.
- **Added:** buildable Graveyard using /sprites/graveyard.png, one Gravekeeper job per graveyard, and up to 50 memorial records per graveyard; names of villagers appear in green and enemy casualties in red. Adult relatives can occasionally visit recorded family graves outside working hours.
- **Changed:** raid outcomes report spoils, own and enemy casualties, village victory announcements or mourning, and leadership rewards in the game Chronicle.
- **Fixed:** the Leader's House can be upgraded, with each level increasing the number of available beds.
- **Fixed:** raid chronicles distinguish threats, marches, combat outcomes, casualties and cancellations; raid totals and village traits no longer count the same raid multiple times.
- **Fixed: pregnancies conceived through youth love keep their origin through breakups, birthdays and Moon Howler transformations; babies take the mother's surname without a bastard label. Existing custody and housing rules remain unchanged.**
- **Added: a read-only weekly village newspaper in the Log window, with real stories, social news and weekly event counts from the existing Chronicle.**
- **Changed: the valley's ecology ladder is live again** — Stable, Strained, Damaged and Collapse now follow the deer, the wolves, the meadows and the town's footprint, thinning hunts and farms and costing reputation, with every warning arriving before its effect.
- **Fixed: the Wildlife Preserve can be built again** — it was left out of the build menu's categories, so its ecosystem bonus and the Keeper of the Wild chapter were unreachable in normal play.

- **Fixed: the shortcut sheet and the Guide now name every window the way the header does** — the doors read People and More instead of Village and Menu, and both say that V and O open the same People window.
- **Fixed: a build spree no longer floods the chronicle** — a day that finishes several buildings of one kind now writes one line with a count ("Wall completed ×30") instead of one line each.
- **Fixed: a festival name that already ended in "Festival" said it twice** — the chronicle read "Harvest Festival festival began in the village"; both festival writers now name the festival once.
- **Changed: auto-assign is now the default for every workplace** — a building staffs itself automatically unless the player sets it to manual, and the manual toggle still keeps a hand-picked crew out of the auto pass.
- **Fixed: an unaffordable building tile no longer accepts a click** — the catalogue re-derived the cost rule locally and used it only for dimming, so an unpayable building could be selected and then refused at the map; it now reads the cost owner and names what is short.
- **Fixed: the long-run gate's checkpoint verdict no longer counts a rival's buildings as the colony's own** — a checkpoint read healthy because a neighbour camp had raised buildings, while the end-of-run verdict correctly saw the player had built nothing.
- **Fixed: the village head's next-vote year no longer prints a fraction** — the office badge read "next vote Y2.2472222222222222" where the election date is a fractional year; it now shows the whole year.
- **Spread the colony's residence assignment across the day** — the day boundary rehoused every settler at once and could freeze the game for a minute; each pulse now sweeps a quarter of the colony instead of the whole at once, so no single tick carries the day's housing work.
- **Made a raid grow with the colony it is aimed at** — the war-band and the loot it is after now scale with the settlers you can muster, so a large village faces a fight worth answering instead of a walkover.
- **Stopped auto-play ignoring a producer nobody works, and let it build ahead of a growing village** — a quarry with no settler on it made no stone while the bot treated stone as handled; it now mans it or adds another, and housing, food, wood, stone and iron keep pace with the population.
- **Bounded that extra spending so it cannot starve the colony** — building ahead now leaves the price of a repair and two days of food in store, while a real shortage still spends the last of them.
- **Let the colony reach its storage buildings** — a silo waited for a full gold store although it holds food, and the barn, silo and wood storehouse sat behind a food reserve a full shelf never needed.
- **Smoothed movement and fixed hover tinting** — settlers, animals and particles blend between ticks instead of jumping once a second (a teleport is drawn straight), building hover tints update every frame and while paused, and all motion runs on one clock.
- **Made a building's hover highlight appear at once, and while paused** — the highlight left the entity layer's cache key when that layer was memoised, so the ring only redrew on the next tick; it is painted per frame now, beside the placement ghost.
- **Put the home and workplace rings on the building they mark** — both were built from the sprite's base anchor as if it were a centre, so a settler's house ring sat half a footprint above the house.
- **Let a vacant leader's house pass to the next household** — the manor stayed with the departed head's family until a new head was elected, and a save preserved the squat; a house is vacated whenever no living settler holds the office.
- **Stopped a prison counting its prisoners as guards against the guard cap** — the workers line measured everyone in the building against the post count, so a fully guarded prison with cells to spare said "No idle settlers"; both numbers now come from the staffing owner.

- **Made the placement ghost and the placed building stand on the same ground** — the ghost sampled the bottom centre of a footprint while a placed building sampled a corner half a footprint away, so a building jumped vertically on a slope.
- **Showed a catalogue tile with no sprite art as the building's own emoji** — a decor type with no image asked the server for the page itself.
- **Made the long-run gate check housing at every checkpoint** — an over-capacity settlement was only tested at year end, so a failure named the house but not the day it first appeared.
- **Read the ecosystem score through its owner** — the village-health line read the raw field, so a legacy save scored an undefined ecosystem.
- **Stopped rebuilding per-frame work the frame did not change** — the ground-chunk plan, the overlay vignette's gradient and the entity-layer cache key are reused until an input actually moves.
- **Kept the blacksmith scan out of camera-only frames** — a panning camera re-scanned every building for a colony-wide fact only a tick can change.
- **Gave a building without a sprite its own footprint pad** — an unloaded sprite drew one white box instead of the building's own category colour.
- **Let a disaster end when hostile weather is switched off** — the feature's early return sat above the loop that ages disasters, so a save carrying one kept it forever and permanently penalised the village's election score.
- **Made the grid markers and zone hints ride the terrain relief** — they were drawn at the flat-map height while the buildings that land on them were lifted, so the two disagreed by up to a relief cap on a slope.
- **Showed the village why a workplace cannot be staffed** — switching a building to manual with no idle settler printed a bare "No idle settlers"; it now lists the settlers the staffing owner refuses and its own wording for each.
- **Stopped a loaded valley growing its forest back over the houses** — a compact save regenerates the ground from its seed, and the record of terrain the player cleared was never written, so every building's footprint returned to forest on load; the sparse overrides now round-trip.
- **Stopped a corrupted save poisoning every later spawn** — a payload missing an id allocator gave every new settler and building an id of NaN, and a rewound one collided with live ids; each is now raised past the highest id the world holds.
- **Made resuming a pause only a resume** — it also auto-staffed every idle settler, so pressing Space undid a deliberate choice to leave a building unstaffed.
- **Taught auto-play to finish the village and raise its buildings** — the bot now builds the well, tavern, workshop, school, church, storage, mill, hotel and taming post once the settlement needs one, and upgrades completed buildings from surplus, only after food, housing and defence.
- **Stopped a silent worker command freezing the village** — a command whose reply never arrived held the tick pipeline shut, and with no tick left in flight the stall watchdog could not fire, so the world stopped advancing while the map kept drawing at full rate. A command now has a deadline.
- **Stopped a leaderless village repeating the postponement every day** — a vacancy that found nobody eligible retried, and logged, once a day; it now re-arms on the same delay the term election uses.
- **Deleted bullshit about buildingid names that is not correct** , residencyOccupancy.hasWorkAssignment=homeBuildingId != null`) is bullshit and made up in relationship file. removed the comments.
- **The Prison now seats the guard crew its own shifts require** — coverage divides the day into three 8-hour shifts, but the building allowed only two workers, so outside hours were certain and every prisoner escaped; it holds four, and the prisoner cap follows the crew.
- **Kept 6 000 chronicle entries instead of 2 000** — a mature village filled the old window with ~200 days of history, so a first-year leader death had already rolled out of the log before anyone could look for it.

- **Capped close friendships at three per settler** — feuds always had a ceiling and friendships never did, so one measured settler ended up close to 54 people and the village read as "everyone became friends"; bonds an old save already holds are left untouched.
- **Gave the village overview a keyboard route** — it had exactly one door, an unlabelled header icon with no fallback, so C toggles it beside O for the citizen overview.
- **Disabled the village portrait panel, which duplicated itself** — its window printed its own title twice, reprinted trait blurbs another window already shows with scores, and closed with commentary about the game's win conditions rather than the settlement.
- **Restored the building panel's inline sections** — its four subjects are visible again instead of an index you had to click into. The header icons are unchanged.
- **Fixed stale election cadence wording** — two places still claimed a decade for a two-year term, including the villager line that said "Decennial year!".
- **Fixed: the idle crowd — settlers go home, to a venue, or somewhere pretty instead of piling into the map centre** — reported from play as *"they again in a circle"* , and identified only because the owner ruled out the two obvious suspects themselves (*"for what there is no election"* , *"i tgink there was no festival"* , *"i got 5 taverns"*). **Three separate rules were stacking settlers on one coordinate, and none of them needed an event.** (1) **The map centre.** Two idle rules — the partner-seeking drift and the idle-leisure wander in humanTick — sent every unoccupied settler to width * 0.5, height * 0.5 inside a 35-slot lattice of offsets, so a village with nothing on piled its whole idle population into one chatting blob in the middle of the map. That is the clump: no election, no festival, five taverns and still one crowd. idleDestination (in the leisure owner) now gives each settler a real destination, chosen deterministically per person per day (personDayRoll, no RNG): **home** ("spending at home"), the **nearest venue** — tavern, market, town hall or church — or **somewhere pretty** off the beauty grid, which is the owner's own list: *"they should goto a tree or to the tavern etc or spending at home various options"* . (2) **The venue point.** The leisure gathering pointed every visitor at a single coordinate — b.x + b.width / 2, b.y + b.height * 0.92, the corner form applied to a value that already is the footprint centre — and each stopped within 20 px of it. venueGatherPosition now gives each settler a slot on three arcs in front of the building, cleared of the wall at every angle, and the arrival test measures the point the settler actually walks to, so the check and the walk cannot disagree (the failure mode the handover recorded for the home/work pair). (3) **The festival.** Its crowd used ((id % 7) - 3) * 11 × ((Math.floor(id / 7) % 5) - 2) * 9 — **35 repeating slots** around the hall or the performers' camp — so a village stacked tens deep on each one; crowdGatherPosition spreads twelve staggered rings of sixteen places instead. **And the ceremony, which the radius cap had made worse:** capping the ring radius without capping attendance meant a thousand settlers shared sixty slots and stood in a *tighter* disc than the wide rings it replaced — attendance is now CEREMONY_ATTENDEE_LIMIT (two rings, 24 settlers, the people the vote is about) and everyone else has a day's work to do. **Measured, not asserted:** 200 settlers now occupy ≥150 distinct places in a festival crowd and 40 settlers ≥20 places at a venue, both deterministic. **Also fixed while measuring:** the venue lookup I first wrote scanned every building for every idle settler on every tick — a hundred thousand comparisons per tick in a thousand-settler village on a loop already fighting for 10× — so the village's venue list is gathered once per tick beside the other per-tick lists. **Reported and cleared in the same breath:** *"the time fast forward doesnt work no anymre"* — measured working from both doors in a fresh build (the 10× button crossed a colony day in under 4 s and the 5 hotkey advanced 01:00 → 16:00 in 4 s), and the owner confirmed it resolved; the tick-cost-at-scale measurement I had started for it was stopped and deleted at their word. **Verified:** humanMovement.test.ts 5 tests (2 new — venue dispersion and the festival crowd), tsc exit 0 on both projects, oxlint 0 warnings / 0 errors on every touched file, full gate **245 files / 1495 passed / 2 skipped / 0 failed** .
- **New: the citizen inspector is an identity plus subjects in their own windows — the "one long stack" the owner reported first** — the last panel on the plan's list (SelectedEntityPanel, the right-hand column). The owner's two statements set its shape between them: the ruling *"each subject should just have its own window not stacking up"* and their own rule for this column, *"right side only citizin information when you click on them"* . So the **right column keeps the identity and what she is doing right now** — portrait, name, Citizen #id, the live activity line, the schedule/target/observed block and

the movement explanation, which is the "why is she doing that" information a player selects a settler for — and her **facts became subjects** : **Life & bonds** (energy, age, hunting/combat, village gear, home, imprisonment, work, clothing, occupation, job skill, pregnancy, and the relationship lines: spouse, secret affair and its build-up, children, sweethearts, courtship) and **Traits** , whose chips used to sit under the name as if they were a subtitle. Each opens in the shared SubjectWindow shell; the family tree was already its own window, and the taming controls stay inline because they are an action, not a fact. **One honest limitation:** the life facts and the relationship lines sit inside the *same* conditional fragment in the original markup, so splitting them into two windows would mean restructuring that fragment rather than re-homing it — they are one subject ("Life & bonds") here rather than two, and that is stated rather than hidden. **Verified in a real browser** (tmp/probe-citizen-subjects.mjs, boot → roster → select a settler): the right column shows the identity with the index, and *Life & bonds* (416×215) and *Traits* (416×210) each open their own window inside a 1600×900 viewport, neither full screen, with Escape closing each and **0 console errors** . **With this, every panel of the sweep has a direct browser look** — the Village tab, ProgressTabPanel, this citizen inspector, and SelectedBuildingPanel, whose own probe places a House through the game's real build path and reads its index (Overview · Construction · Actions · Advanced) and *"House — Actions"* opening at 416×75. tsc clean on both projects, oxlint 0/0, and the full gate npm run test:standard green at **245 files / 1493 passed / 2 skipped / 0 failed** .

- **Fixed: the election is findable in the Chronicle, and the gathering is a crowd beside the hall instead of a ring across the map** — two follow-ups from the same report. **(1) Its own event type.** The owner: *"an i didnt get a mesage about a election ceremony"* , then *"there is nothing in the logs about it"* and *"2000 lins of logs is quite some time so a electrion should be visibel"* — and they are right that a ceremony line should survive the 2 000-entry window, because the problem was never the window: every leadership line was written as **'event'** , the bucket that held **1 531 of those 2 000 entries** in the earlier audit, so an election could not be filtered out of it. There is now a real **'election'** type with its own **Elections & leadership** filter, a icon and colour, and all ten leadership lines write it (the founding appointment, a postponement, the ceremony beginning, the revelry, a failed reveal, a no-candidate year, elected, succeeded, and "can no longer lead"). **Deliberately not applied to electionPromises' lines:** legacyGoals.findPromiseVerdict reads the campaign-promise verdict by type === 'event', so retyping those would have silently ended that achievement. The compiler named all three sites that had to agree — EventLogPanel's icon and colour Records and tests/eventLogPanel.filters.test.ts's exhaustive coverage Record, the guard the imprisonments fix added — which is exactly what that guard is for. **(2) The ring no longer grows without bound.** The owner: *"and why not all are in the circle?"* . Because ringRadius = 22 + ring × 14 had no ceiling: a village of 300 eligible adults was sent to slots up to **358 px** from the hall, and 600 adults to **708 px** . At the ceremony's own walk speed (base speed × the 1.15 humanTick applies, ≈ 2.3 px/tick) that is ~300 ticks of walking against a **70-tick** ceremony, so most of the village was still strung out along the way when it ended — which is what the owner was looking at. Rings are now capped at MAX_GATHER_RING = 4 (**78 px**), with the overflow sharing the outermost rim on their own stable angles and non-attendees just outside it; the ring maths for slots inside the cap is byte-identical, so tests/medium-N5-electionGatherSlot.test.ts still pins it unchanged (its 15-attendee fixture reaches ring 1 and outer ring 2, both inside the cap). Measured on a 300-adult village (tmp/ceremony-crowd.mts): furthest slot **358 → 78 px** , and the share of attendees that can physically reach their slot inside the ceremony **26 % → 76 %** (simulated arrival **82.3 %** , 247/300; the rest are up to 56 px short when it ends). **Stated plainly:** a one-day ceremony cannot assemble a village spread over a kilometre at settler walking speed — the remaining ~20 % is that physics, not a placement bug. tsc clean on both projects, oxlint 0/0, full gate green.
- **Fixed: the election ceremony is one day, not two in-game months — and it now tells you it started** — reported from play while the village stood still in a ring around the Town Hall: *"they are in a circle"* , *"they not walking"* , *"but a ceromeny should maybe take one day?"* , *"not months"* , and *"an i didnt get a mesage about a election ceremony"* . **Two separate defects, both found by reading the ceremony's own owner. (1) The duration.** getPhaseTicks('gossip') was GOSSIP_MONTHS = 2 **in-game months** (2 × DAYS_PER_MONTH × TICKS_PER_DAY), and humanTick.ts:722 holds **every eligible adult** at its ring slot for as long as a ceremony is live — vx/vy zeroed inside 10 px, suppressIdle and onSchedule both set — so the whole village stopped walking, working and eating for a fifth of a year. Nothing needed the hold: the gossip *lines* are delivered by tickElectionGossip on its own cadence from tickElectionCeremony, independent of

where anyone stands, and the ceremony's rings are untouched (that maths is pinned by tests/medium-N5-electionGatherSlot.test.ts). The phases are now named in **hours** — gathering 12, gossip 10, tension 1 — and the ring's own comment records why: an election is a once-a-decade event, so a day of assembly *is* the ceremony. Measured end to end (tmp/ceremony-length.mts): **70 ticks = 23.3 in-game hours = 0.97 of a day** , phase order gathering → gossip → tension → reveal, ending with its New village head announcement and clearing the ceremony. **(2) The silence.** Starting the ceremony wrote a **Chronicle line only** (logEvent(state, 'event', …)), so it landed in the generic Events bucket with no card, no feed entry and nothing on screen — while the *buildup* a year earlier did raise a notice, which is exactly why the absence read as a bug. The start now also raises a real notification — *" Election ceremony begins — the village gathers at the Town Hall to choose its head (one day of assembly)"* — carrying the focus point addNotification supports, so the entry is clickable onto the gathering site, the same door the priority alerts use. The gossip status line dropped its stale *"Months of gossip"* wording with it (villageLeadership.ts), and the now-unused GOSSIP_MONTHS/DAYS_PER_MONTH/TICKS_PER_DAY constants are deleted rather than left as a second statement of the calendar. tsc clean, oxlint 0/0.

- **New: the Progress panel's Goals tab is an index of six subjects, each opening its own window — the last measured stack in the game** — ProgressTabPanel still held the six CollapsibleSections the plan counted (four inside GoalsPortraitPanel, two in the goals branch), and it now follows the same shape as the Village and building panels: one **index** — How history sees you · Your path (live) · Challenges · Valley statistics · Valley chronicle · Dynasties, each with its glyph and a one-line hint — and one SubjectWindow per subject through the shared shell. PROGRESS_SUBJECTS is the single owner of that list, so a subject cannot be listed without a window behind it, and GoalsPortraitPanel now supplies only the four bodies plus the portrait computation they share, because the index is one list in one place. **Deliberately unchanged:** the Research and Trade sub-tabs keep their existing inline lists (they were never the measured stack) and the sub-nav still selects the three groups; only the Goals tab became an index. **Verified in a real browser** (tmp/probe-progress-index.mjs): the Goals sub-tab renders all six index buttons, *Challenges* opens at 416×681 and *Valley chronicle* at 416×681 inside a 1600×900 viewport — neither full screen — with 0 console errors.
- **Investigated and deliberately not "fixed": a worker slot can overlap a building footprint, and that is not a reachability defect** — the per-worker row introduced above is wider than a small building (eight workers on a 40 px farm span 112 px), so in a dense layout **17 of 33 slots** landed inside *some* building's footprint (measured with tmp/are-worker-slots-reachable.mts). That looks alarming and is not: pathfinding.buildPathGrid blocks **only BuildingType.Wall** (isBlockingWall, pathfinding.ts:44), so a slot on a farm's footprint is walkable and the settler arrives. A slot inside a **wall** footprint would be unreachable, and that exposure is not new — the replaced single-point formula had it too, since its y = building.y + h × 0.92 also lands in the footprint directly below in a dense column. Recorded rather than papered over with a rule nothing has measured. Separately, the owner's *"they not walking"* report was measured against this tree before touching anything: a fresh colony with the whole map in view moved settlers in **85.0 %** of settler-ticks at a mean **0.686 px/tick** (npx tsx scripts/probe-move-distance.mts 3 focus), so the movement path is healthy and the destination change alters *where* a crew stands, not whether it walks.
- **Fixed: a building's crew stands in a row in front of it instead of clumping into one spot — the workplace stand point was half a footprint off and spread by a 7-value hash** — owner, looking at a Hunting Spot crew: *"why are they all standing at side of the village?"* , then *"but why all at the same palce not logic"* . Both halves of that were literally true in one expression, humanMovement.humanBuildingTarget's workplace branch: **(1)** the point was built in the **corner form** — building.x + building.width / 2 — on a value that already *is* the footprint centre (buildingGeometry.ts's own header records this form as the repo-wide trap, 86 sites at the last count), so a crew stood half a footprint to the right of its own workplace; and **(2)** the only dispersion was ((|entityId × 13 + buildingId × 29| % 7) − 3) × 6: **seven** possible x positions in a ±18 px band, an **identical y for every worker** , and past seven workers the hash collides so two settlers stand on the exact same pixel — which is what "all at the same place" means, and why it read as illogical rather than as crowding. The slot now comes from the list that owns the crew (building.occupants, with the ?? [] its other save-boundary readers use) and the row is centred on the footprint, so the geometry is deterministic and cannot collide. **Measured, not asserted** (tmp/where-do-workers-stand.mts, an 8-strong crew on a 40×40 spot): before — **7 distinct positions of 8** , a **36 px** band,

row centre **20 px right** of the footprint centre, y at h × 0.92; after — **8 of 8 distinct** , a **112 px** row ((crew − 1) × WORKER_STAND_SPACING_PX), centre **exactly on** the footprint centre (Δ 0), standing WORKER_STAND_STANDOFF_PX (10 px) south of the bottom edge, i.e. in front rather than beside. **One pinned test recalibrated, not weakened:** tests/medium-N4-campAnchor.test.ts pins the barracks guard's position after a tick, and that guard's post moved (one worker: x + 15 + offset, y + 27.6 → x, y + 25), so its velocity differs in the fifth decimal; the case's *invariant* half — the settler average moves inside the tick and the anchor follows the live average — is untouched, and the new numbers carry a comment saying they move only with the destination owner. **The same class of bug is deliberately left in place elsewhere:** homeStandPosition's porch arc and the leisure/venue stand points (humanLeisureBehavior, humanTick) still use the corner form, because the night-hiding predicate (isNearResidence) shares that point and the handover already records a regression from fixing one side of that pair alone (49.5 % → 39.8 % hidden).

- **New: the Village panel is an index of six subjects, each opening its own window — the last stacked panel in the game** — the owner's ruling (*"each subject should just have its own window not stacking up"* , *"i dont want any thing stacked"*) applied to VillageTabPanel, which was still seven CollapsibleSections scrolling in one column. It is now an **index** — Population, Food this day, Village leadership, Household roster, Armament, Reputation, each a button with its glyph and a one-line hint — and each subject opens in its own window. Two placements are deliberate rather than incidental: **HousingDiagnostics moved inside the Population window** (housing is that subject's business; it used to sit between two others in the stack, and its own disclosure inside the window is the progressive disclosure the plan allows *inside* a subject), and **the reputation explainer was promoted to its own subject** so that nothing at all is left stacked in the tab. FocusPanel stays inline — it is the tab's "what to do next" action strip with its Go → buttons, not a data subject — and that is the one thing here still rendered in the panel rather than a window. **New shared components/SubjectWindow.tsx** : the wrapper is extracted rather than copied because SelectedBuildingPanel had grown an identical one, and a second copy is exactly the mechanical clone this repo's jscpd gate fails on; it owns the shell, the keyboard claim (useOverlayKeyboard, so v/f/Esc cannot act on the map behind a modal opened from a panel) and the rule that a closed window mounts nothing, while GameWindow keeps the focus half. **Verified in a real browser** (tmp/probe-village-index.mjs, reusing the repo's CDP harness): the Village window carries all six index buttons and no stacked section headers, and clicking one opens **that subject's own window** — Food this day 416×110, Household roster 416×375, both inside a 1600×900 viewport with real margins, neither full screen — with Escape closing each and **0 console errors** . tsc clean on both projects, oxlint 0/0, and the full gate npm run test:standard green at **245 files / 1493 passed / 2 skipped / 0 failed** .
- **New: every overview subject is its own window with its own icon in the header — the six-panel full-screen Valley overview is gone** — owner: *"the 6 panel stacked now that opens full screen should be each subject a own icon the header and open a window for that subject not full screen"* , then *"just a icon not a name with it in the ehader"* , then *"the header is now quite large 3rows"* . CitizenOverviewScreen was the last full-screen takeover: a fixed inset-0 dialog with its own header, a six-chip nav strip, six status cards and the panel below them, all covering the map. It now renders into **GameWindow** (centred, 928 px wide, body capped at 70 vh) and shows **one subject** , with the nav strip **deleted** — the way to another subject is its own icon in the header, not a chip inside a panel. **The header gained five doors** (⚔ Frontier, 🌿 Nature, ⚙ Progress, 📜 Log, ☰ More); the sixth, **Village** , is the 👥 population chip that already opened that subject, so all six have a door without the bar carrying two glyphs for one thing. Each is **icon only** per the owner's second report, with the name in the title and the accessible name, and each is wired through the existing route owner (sidebarTabForOverviewNav) rather than a second list, so a subject cannot be reachable from v/f/n/p/l/m and missing from the header. **One owner for the subject's identity:** new OVERVIEW_SUBJECTS in hooks/useGameShe…4050 tokens truncated… centre — it is the anchor the sprite is drawn from so the art rises off the footprint base, a convention both buildPreview.ts:23andgrid.ts:331record. The map-ring code was the one caller that ignored it, subtractingw/2, h/2as if it were a centre. The ring is now built from the base: horizontally centred onsx, extending **upward** from sy`. This was the "selecting a settler rings their home (sky-blue) and workplace (amber)" feature from the previous session's handover, which had never been looked at in a browser.
- **Fixed: the family-tree button existed only for settlers who already had relatives, so for most of a young colony it was not there at all** — reported from play as *"it doesnt work when i cick on famliy three"* . The inspector rendered its Family

tree door behind tree.parents.length + tree.siblings.length + tree.children.length > 0 || tree.partner != null (SelectedEntityPanel.tsx:569), so a settler with no recorded parent, sibling, child or partner — which is most of a freshly-founded village — showed **no control** , and the feature read as broken rather than empty. Worse, FamilyTreeWindow already had an explicit empty state (*"No living relatives recorded"*) that the condition made **unreachable** . The door is now always rendered and labels the case on the right (no relatives recorded), so the window says it instead of the button silently vanishing. **Verified in a real browser** through the canonical bot, which now drives the whole chain: roster → Locate <name> on the map → inspector → the tree. Result: opened: "Family tree of Jere" close="Close Family tree of Jere", closed / gone: {"ok":true} / true, 0 console errors. Two tooling fixes were needed to get there and are worth recording: every earlier selector attempt *guessed* at the DOM and failed, while the working one was read from the source (PopulationPanel.tsx:57); and matching the opener by its visible text hit the *collapsible section header* Family tree instead of the button, so it now matches the button's title.

- **Fixed: the "Family tree" was not a family tree — it was an alphabetical surname list, and it was open by default** — reported from play as *"family three is not a family thee"* , and the screenshot confirms it: the Village tab rendered a defaultOpen CollapsibleSection titled **Family tree** whose contents were families as flat groups — Musson with 2 adults, Mace with 2 adults, Wing, Andersen, Beeler… one row per surname. It was both **mislabelled** (a list of surnames is not a tree) and the worst instance of the stacking the owner rejected in the same breath (*"i dont want any thing stacked"*), because being open by default it dominated the tab. It is **deleted** , with the reason recorded where it stood so nobody re-adds a list under that name. The real tree (FamilyTreeWindow, opened from a selected settler) now **draws** one: parents across the top, the settler with their spouse as the anchor row, siblings hanging off the same parents, children below each on its own stem with a connector — so descent is visible rather than inferred from section headings. That is the difference between the earlier "grouped by generation" pass and this one: grouping is still a list of groups, and a tree needs the line.
- **New: GameWindow, the one window shell every subject renders into, and the work & venue hours editors move into their own window** — owner ruling: *"each subject should just have its own window not stacking up"* . components/GameWindow.tsx is the primitive — title bar, one close control, one scrollable body — and owns **no** subject logic, so adding a window is a small change rather than a new layout. FamilyTreeWindow was refactored onto it (it had hand-rolled its own chrome, so the shell did not exist yet), and the three schedule editors that were buried inside CitizenOverviewScreen now live in components/WorkHoursWindow.tsx, opened from the Village section by an **Edit hours** button. The editors themselves are unchanged owner-owned panels, so the Apply button behaves exactly as before; routing them into the shell is the whole change. **Deliberately not draggable yet** , and stated in the component: the plan lists draggable/resizable as part of the full shell, there is no drag or resize helper anywhere in this repository, and that is new interaction work rather than a refactor — every window is anchored top-right until it lands.
- **Changed: the work and venue schedules have no length restriction — the owner sets how long the colony works and how long the Tavern and Hotel stay open, and a long window is staffed rather than refused** — reported from play as *"i cannot change work times"* and *"i choose how long the tavern should be open"* . The panel was not broken: a 04:00–23:00 (19 h) Tavern was being **refused** by MAX_VENUE_SERVICE_HOURS = 18, and the ordinary work window carried a 2–16 h band. Measured on the owner's own screenshot, the status line read *"Blocked: Venue service cannot exceed 18 hours"* with the Apply button correctly disabled — a bound, not a bug. MAX_VENUE_SERVICE_HOURS and MIN_VENUE_SERVICE_HOURS are deleted, and so are MIN_STANDARD_WORK_HOURS (2) and MAX_STANDARD_WORK_HOURS (16); the only width limit left is the long-standing **no-wrapping-past-midnight** rule, so 1 hour is the narrowest window and 23 hours (00:00–23:00) the widest. Verified in a real browser, not by reading: the ordinary work panel accepts **00:00–23:00 (23 h)** , enables Apply, commits it, and the current line reads 00:00–23:00 (23h); the Tavern accepts **04:00–23:00** and commits it. **The staffing rule is unchanged and is the point of removing the cap** (owner: *"if its goes over 8 hrrs a extra person should go work at the place"* , clarified *"well 9 hrs also fin"*): getVenueAutoStaffingTarget keeps ceil(hours / 9) bounded by the building's own crew cap, so 1–9 h is one settler, 10–18 h two and 19 h+ three. A venue open all day is covered by more staff, never refused.

- ***Fixed: production is paid for by settlers who actually turn up — output scaled with the configured work window while a settler asleep in bed counted as working*** — reported from play as *"is work hours related to production at this moment and i mean the person shoud eb really at the work palce"* , and the report was right on both halves. **(1)** dailyBuildingEconomy and workshopEconomy both derived their multiplier from getWorkScheduleHours(getWorkSchedule(state)) — the *configured window* , not work done — so setting a 23-hour day multiplied every building's output by **2.56×** without anyone being at work. **(2)** The only presence test anywhere in the production path was staffed = workers > 0, and workers came from buildingWorkerStats, keyed on worker.homeBuildingId — an **assignment** . Nothing read an entity's position, so a farm produced its full 12 + workers × 5 while its worker slept at home. Attendance is now measured and shared: resolveDailyScheduleFatigue snapshots the hours each settler actually worked into scheduleLastWorkedHours **before** zeroing the tick accumulator (it runs at tickLayerDaily:42, ahead of the economy at :81, so without the snapshot the only record of attendance was destroyed before production could read it), and one exported rule — getWorkplacePresenceShare(workedHours, assigned, scheduleHours, hasAttendance) — is applied by real production **and** by the workshop estimate, so a preview cannot drift from output. A crew that works its whole window scores **1** and keeps exactly the output the window alone used to give; a crew that never turns up produces **0** ; an unmeasured world (a fresh load, before the first day boundary) reports 1 rather than 0, because "not yet measured" must not read as "nobody ever works".
- **New: a separate testing bot that raises the whole building catalogue — and it proves a settler is physically at the church, not merely assigned to it** — owner: *"give the bot just options to build all things so cheating but fo rtesting"* and *"keep virtual-player.ts apart thats for in game and use own for testing"* . scripts/test-bot-build-all.mts drives the real command boundary and **closes a recorded coverage gap** (BUG_REPORTS/2026-09-29-automated-runs-never-build-a-church.md): no automated run in this repository had ever reached churchStrength > 0. Result: **21 of 34 buildable types raised, the world still simulates, and the church is live at strength 1.** Two traps were found by running it, both recorded in the script: **(1)** buildDebug places a *construction site* , and getChurchStrength is gated on b.completed — so the first run reported "church built: true, churchStrength: 0"; **(2)** countWorkersAtBuilding counts homeBuildingId, i.e. an **assignment, not an arrival** (the same assignment-vs-attendance distinction as the production fix above), so the bot now ticks until the keeper is **physically at the church** before it claims success — owner: *"no it need people workin there"* . Two opt-in primitives support it and are unreachable from any shipping path: a buildDebug worker command and startBuilding(..., free), which skip **research and cost only** — terrain, overlap and one-per-village placement are still enforced by their owner, because two buildings in one footprint would corrupt the spatial indices rather than merely cheat. scripts/autoplay-browser-probe.mjs also gains a reusable CAPABILITIES registry, so the UI-driving bot gains named handles onto the game's screens instead of each test re-deriving selectors.
- **Investigated and deliberately reverted: the in-game bot does not build a Church, and the reason is its own pinned ordering** — an attempt to extend the shipping auto-play ladder (owner: *"do extend its building options so let it build a church and other things"*) failed **twice** , at the civic step (**36 of 68** cases in tests/virtualPlayer.test.ts) and at the end of the ladder (**15 of 68**), and pre-placing a church in the shared fixture made it *worse* (**18**) because the Church is itself a manual-staff workplace. The cause is not a bug: those 68 cases pin the exact decision sequence and treat null ("the colony needs nothing") as a reachable healthy state, and any step that fires in a small, well-stocked fixture contradicts 15 terminal expectations. Extending the in-game repertoire is therefore a redesign of the bot's identity — **does a healthy colony keep building?** — and that is an owner decision, not an edit. The ladder is left exactly as it was and virtualPlayer.test.ts is green at 68/68; the church coverage gap is closed by the separate testing bot above instead.

⚔️ Warfare, Raids & Combat Resolution

- **Added:** a paced full-screen text chronicle for outgoing raid battles, presenting successive phases and recorded battle outcomes and named casualties.
- **Changed:** victorious outgoing raids cause permanent rival population losses, with overwhelmingly successful attacks able to eliminate rival settlements and clear associated events and structures. Rival deaths are recorded individually where characters exist, alongside counts for off-map casualties.
- **Changed:** combat CSV exports include raid stage and kind, and the combat panel labels each stage.
- **Changed: wolf fight-backs at the Hunting Spot now bite in one hunt in ten rather than one in three** — and a hunter's own trade skill cuts the fatal band from 5% to 1% at mastery.
- **Fixed: the Wildkin is now the apex creature it was meant to be** — it ate grass and its bites were billed against the pasture, so the top of the ladder read as a deer; it now eats meat, is no longer a grazer, and its birth, first sighting and death are chronicled with the place each happened.
- **Changed: grass and trees no longer carry a combat roll seed** — only the species that take part in a predator or prey roll are seeded, which trims a number a blade or a trunk can never use and shrinks every save.
- **Opened the frontier in time to matter** — a neighbour camp now appears once your colony is real instead of waiting years on a rare event, so raids, the raid banner and the raid action can exist in a normal run.
- **Made rival camps roll for a raid every day** — the roll only ran on the camp's own action day, once every two to three months, so raiding could go a whole game without happening; the camp's own cooldown and odds now set the pace.
- **Gave a refused raid the reason the rules already return** — marching on a rival re-implemented six eligibility gates and returned silently, so a blocked raid button did nothing and explained nothing; it now shows the same reason the eligibility owner gives.
- **Stopped the dashboard and the combat log freezing at the values they opened with** — both memoised a projection on the world object, which the simulation mutates in place, so neither could see a change while the simulation ran on the main thread.
- **Reached the chronicle and the combat log pre-filtered from a concern** — the dashboard's concerns now open the log on their own filter.
- **Gave blocked trade and raid buttons a visible reason** — the refusal lived only in a hover tooltip, which is unreachable on touch and on a disabled, unfocusable button.
- **Made a wolf able to kill** — a fight-back could only ever cost the hunter energy, so a long colony's chronicle held thousands of bites and no deaths; one roll now chooses between killed, mauled and an ordinary bite at the owner's odds.
- **A jailing now raises a notice that waits for you** — an imprisonment was announced only as one of the four transient toasts, so a rare jailing could pass unseen; it now posts a card like a raid or a festival does.
- **Changed: the Hunting Spot's hunter walks to the animal, and only takes it at arm's length — half of this is verified working, the other half is not yet** — from the owner's report *"why is it so difficult that hunter goes to the prey"* , the ruling on reach, *"120 px is way to far, its 1800's they dont have guns"* , the reminder *"dont forget animals fight back"* , and finally *"let the animals just dont run anyway"* . **What was there:** the assigned hunter stood at the spot's work point and the shot resolved only if an animal happened to wander inside **120 px** (~10 m) of it — a turret, not a hunter. The only settlers who walked to prey were *job hunters* free-roaming, and only when below 85 % energy, which is why a staffed spot looked inert. **What changed, in the owner's order:** (1) **a move-to-prey order** — huntingSpotChaseTarget (new owner module src/game/huntingSpot.ts) is applied in humanTick's *work-commute* branch, so the assigned hunter walks at the animal and holds beside it. That branch, not the shift branch above it, is the one a spot's worker actually runs in: measured, a hunter stood frozen at its post (v=(0,0)) all day while its quarry walked from 193 px to 209 px. (2) **Kill reach is bodies touching** — huntingKillReach = hunter radius + prey radius (≈21 px against a deer, ≈17 px against a rabbit), the same expression humanHuntingBehavior uses and the same rule the wolf mauls on. The 120 px constant is deleted, and with

it the note that predicted this change: *"the honest fix is the chase"* . (3) **The fight-back is unchanged and now lands on the hunter** — same 35 % wolf roll, same 85 % accuracy roll beside it; only the victim moved from the Hunting Spot's health (an animal the settler was never near, hit from up to 320 px) to the settler standing next to it, since a settler's health *is* its energy. (4) **The hunted animal no longer flees the settler hunting it** — isWildlifePredator counts every adult settler, so a grazer inside its fleeRange bolts at speed × 1.5: a deer at **4.5 px/tick** against a working hunter's ~3.6, which is why a hunter could never close. The exception uses the index that already exists for exactly this pairing (ctx.huntTargetByPreyId, prey id → the hunters after it), so wild predators and every *other* settler still spook it. (5) Supporting: a spot **commits** to one animal (huntingSpotTargetId) instead of re-picking the nearest each tick (measured: the pick cycled #357 → #431 → #452 in one day and the hunter closed on none), a commitment survives past the eyes' 350 px, the hunter's pursuit is paced (HUNTING_PURSUIT_SPEED_MULT = 1.4) and aims where the animal is going (huntingPursuitPoint), and a contact is recorded (huntingSpotInReachTick) because the production pass samples a single instant. **Verified:** the hunter leaves the post and closes — 0 px → **306 px** from the post, closest approach **65 px → 10 px** once the hunted animal stops fleeing — and the full gate is green (245 files / 1493 passed / 2 skipped / 0 failed), with three fixtures updated to stand the deer beside the hunter because a contact reach is the point of the change (huntingSpot.cleanup, foodLedger.acceptedCatch, low-9-simdata). **NOT verified, and stated plainly: a catch still does not land in the measurement** — 162 of 162 spot production passes over twelve colony days decided "too far to shoot", every one of them in the small hours with the hunter back at its post, and the contact record did not change that count. So the *walking* half of this works and the *taking* half does not yet. The reason is now clear and is a design fork, not a bug hunt: a **job hunter's free-roam path already walks to prey and kills at contact distance** (humanHuntingBehavior, gated on energy below 85 %), so the spot's separate per-pass sampler duplicates a hunt that already exists — the two should be one, with the spot's work *being* the hunt. That unification is the next step and is the owner's call.

⚖️ Elections, Politics & Career Merit

- **Changed:** the three leading election candidates stand at the center of the ceremony, election phases last longer (24 hours gathering, 20 hours discussion, 8 hours final tension), and each phase is recorded in the Chronicle.
- **Changed: each election gets a seeded swing of up to ±20% in candidates' merit support, so a close race can go either way while a large lead still matters.**
- **Changed: one capped trade now adds 50 merit, total skill merit caps at 100, and incumbent record penalties are softened and bounded.**
- **Changed: election merit weighs a whole career, not only trade skill** — experience, Town Hall service, friendships, titles and the incumbent's record all contribute to the result.
- **Refused a corrupt election-buildup year at the save boundary** — the year the buildup notice was raised was the last leadership value accepted without a rule, so a save carrying an unusable one loaded and then misread the notice; it is refused by name now, like its siblings.
- **Stopped panning the map throwing away what you had selected** — the click that follows a drag arrives after the drag ends, so every pan cleared the selection and collapsed the inspector; a drag is now left to the camera.
- **Kept the diagnostics drawer when nothing is selected** — clearing a selection silently removed the panel the player had deliberately opened.
- **Listed every taming candidate instead of four** — the picker silently truncated its list, and now says how many are available and why a button is dead.
- **Let a founding-year election's promises count** — year 0 is a real election year, but promises recorded in it read as "none recorded", so they were never shown and never judged.

- **Announced the election revelry, which ran silent** — the party boosts production, courtship and immigration but only wrote a Chronicle line, so its effects arrived with no word about them; it now raises a notice like the ceremony start does.
- **Gave feuds a real source** — two settlers whose traits clash and who share a home, workplace or job can now drift into one, as can a wrong: a famine foot-bite, losing an election, or a rumour that reaches a cheated spouse. A 500-settler colony settles near a dozen live feuds.
- **Changed the colony's social news to one line a year** — routine friendship wrote a line per bond, which made 60 % of a real 2 000-event chronicle friendship noise and buried the leadership election inside it.

Settler AI, Pathfinding & Life Simulation

- **Fixed:** prison guard patrols retain the assigned crew and do not leave their posts for social walks while commuting to a shift.
- **Changed: children decide once per in-game hour whether to play with a nearby child, then keep steering toward that playmate between checks.**
- **Changed: settlers walk commutes without teleporting or distance-based speed spikes; urgent walks use a fixed speed cap, and stuck settlers recover onto nearby walkable ground.**
- **Fixed: the shift-fatigue line dropped a word** — "Short shifts spared 1 settler that output cost" now reads "spared 1 settler from that output cost".
- **Fixed: childhood now ends at the adult floor** — a settler became an adult and took a job at 12, so children filled the postings while adult settlers sat idle and their children could not be apprenticed.
- **Fixed: a settler widowed mid-pregnancy never married again** — conception marks both spouses as expecting and only the birth clears the father, so a wife's death stranded her husband in a status the courtship rule refuses; existing saves repair on the next tick.
- **Fixed: an on-duty prison guard no longer abandons his post to visit his wife** — a free-time walk to a distant spouse overwrote the work commute in the same tick, so the rota was full, nobody stood on the post, and every such hour rolled a prisoner escape.
- **Fixed: animals and settlers no longer flick back to where they stood a moment ago** — the render snapshot and the simulation were both writing entity positions, and the snapshot is a past tick, so it dragged every mover back one buffer behind itself.
- **Fixed: a settler-only pregnancy counter no longer rides on every blade of grass and tree** — it was initialised on every entity, so each save carried a number only a mother can use; it now appears only once a settler is expecting, which shrinks every save.
- **Changed: grass and trees no longer carry an animation frame or a settler generation** — only species whose animation clock runs, and only settlers in a family tree, read them, so both numbers leave every blade and trunk and the save shrinks.
- **Fixed: settlers no longer walk over water or through walls** — the grid that blocks them was read only on long walks, so a short leg, or one with no way around, crossed the obstacle. Every step now asks it and slides along what it refuses.
- **Fixed: a bridge is now a crossing** — a completed bridge was marked as walkable but left the water under its deck blocked, so the router still refused it and the settler waded the river beside it.
- **Fixed: walkers no longer cut a corner diagonally** — the route advanced to the next waypoint from half a tile away, which cut one-tile gaps and left the walker pressed against the bank.
- **Added: the save inspector audits the ground the colony stands on** — a sent-in save now names anyone standing in water, inside a wall or on reserved decor, and can run the colony forward to show whether anyone walks onto it.

- **Added: the hotel and hospital inspectors name everyone staying there** — the hotel lists visitor guests and settler lodgers with whether they are asleep, and the hospital lists each patient in the ward with the reason they need care.
- **Fixed: a visitor walking to a hotel no longer re-plans the whole route every step** — the route cache key held the visitor's moving position, so it missed every tick, re-ran the pathfinder and churned the shared cache; it is keyed on the visitor now.
- **Fixed: a settler cursed into Moon Howler form mid-romance no longer comes back to a one-sided sweetheart link** — the pair is held still while one is transformed, so the bond stays mutual.
- **Made settlers follow a cached route to its end instead of re-planning every few steps** — a long blocked walk re-ran the pathfinder every 32 px of progress although the route had not changed; it now re-plans only when the walker leaves the route.
- **Made the pathfinder drop one stale route instead of clearing the whole cache** — past 200 settlers pathing at once, the cache emptied and everyone re-planned from scratch; it now evicts only the oldest.
- **Fixed: the fatigue line no longer reports an average of worked hours as a shift** — a fractional day like 22.3h read as an impossible shift length; it now says the settlers worked that many hours on average.
- **Made the long-run gate judge the colony, not only its invariants** — a settlement that raises nothing while its beds are full and settlers stand outside them now fails the run as stalled, and a stalled default scenario is reported rather than the rule softened to pass it.
- **Let settlers sleep at a hotel when home is too far** — a settler whose walk home would outlast the hours before their sleep hour now lodges at the nearest completed hotel instead of crossing the map; the eight beds are shared with paying guests and the cap is never exceeded.
- **Raised a hotel's sleeping capacity from four to eight** — the staffed hotel's beds are one pool, shared by paying visitor guests and by settlers lodging overnight.
- **Kept the night prison guard on his post, and let settlers work through a festival** — a pre-dawn trip home teleported the rostered night guard off his post, so those hours counted unguarded; and a festival suspended everyone's work, where it now gathers only settlers who are off shift.
- **Made a feud's end announced** — a feud could be announced as brewing and then fade with no word, because the start line and the end line used different bars; both use the same one now, and a feud ended by a death is left to the death line.
- **Enlarged the three map sizes to the renderer's full spec** — Medium 2560×1920, Large 4096×3072, Huge 6144×4608; settlers walk twice as fast so a commute still takes the hours it did, the smaller-map assumptions were re-derived, and a big-map colony saves to a file instead of the in-game slot.
- **Repointed a settler-trait note at the modules that read it** — the comment named a simulation file that was split into domain modules long ago.
- **Stopped a house holding more settlers than it has beds** — a marriage that could not seat the whole family, a divorce and a birth each wrote a residence without asking for a free bed, so a six-bed house could hold ten; every such move now takes the residence owner's room rule.
- **Stopped a home with room for one bed being asked to take two movers** — the least-crowded rule discounted a mover who already lived there twice, so a settler marrying someone already housed walked the couple into a full house.
- **Showed a settler's name instead of their internal id** — the citizen panel labelled people with the save's allocator id.
- **Added a gender glyph to settler name plates** — gender was carried by label colour alone.
- **Stopped offering a settler a post they could not walk to** — assignment took the nearest idle settler whatever the distance, so a family housed across the valley could be handed a school they could never reach; a post over two hours' walk from home is now refused, and a road on the route helps.

- **Made the commute estimate agree with the walk it describes** — the lead-schedule estimate omitted the travel multiplier the work commute actually uses, so every reported commute read 3.5× longer than the one walked, and settlers set off for work hours earlier than they needed to.
- **Counted a settler's shift even when the camera was elsewhere** — the off-screen throttle skipped the attendance counter on 23 of every 24 ticks, so a workplace's output depended on where the player was looking; attendance is now recorded for the shift, not for the frame.
- **Moved a docked caravan's wait onto the clock it is measured in** — the countdown ran inside the same per-settler throttle, so a caravan the player watched reached its partner up to 24 times sooner than one they did not.
- **Stopped the yearly death tally counting departures as deaths** — finished caravan carriers and departing visitor groups are despawned by marking them not alive, so every completed trade run added a phantom settler death to the year review.
- **Cleared the relationships a loaded colony cannot resolve** — a spouse or courtship id naming nobody left a settler stuck as married to nobody, who could then never widow and was shut out of courtship permanently; a load now sweeps those ids, reusing the same owner the death path uses.
- **Stopped divorced couples remarrying the next day** — nothing recorded that two settlers had already been married, so a pair who still shared a house courted and remarried immediately after every divorce; a marriage now keeps them apart for a year.
- **Let the affair system actually run** — its daily check compared a tick of the day against an hour, so it could only match the small hours when the settler was asleep at home and its own rules refused it; most married settlers never reached it at all.
- **Made settlers clickable** — the hit box was measured in world units and shrank as the player zoomed in to pick someone out of a crowd; it now has a floor measured on screen.
- **Gave the arrow keys one owner** — they panned the camera and stepped the map cursor at the same time, which made the keyboard route onto the map unusable.
- **Cut the day-rollover freeze from 58 s to 18 s, and it is still going** — residency assignment asked whether a house held a couple, a minor or only singles by re-scanning the whole colony per question, per house, per family; one walk now answers all three.
- **Gave prison guards days off and split the jail's posts from its cells** — the Prison seats five guards beside four cells, three on watch and two resting each day, a resting guard no longer walks to a post he is not rostered for, and prisoners are no longer listed as the jail's workers.
- **Fixed the population benchmark measuring one colony at every tier** — the per-tick maintainer re-pinned the fixture to 300 settlers, so all six tiers ran the same population; each tier is now that fixture at its own size.
- **Drew the family tree instead of listing names** — the window drew one decorative stem per generation, so no name was joined to any other; each parent now branches to their own children, and a settler's card opens the tree from one icon.
- **Named the jobless state honestly and gave the teardown one owner** — the shared job label called an unemployed settler a "Settler" as if it were a trade; it now reads Unemployed, and one helper clears the workplace pointer, job and occupation together.
- **Fixed the daily friendship pass starving large groups and bonding the jobless** — a group larger than the pair budget handed every bond to the same members, and the unemployed counted as sharing a trade; the window rotates daily and only a real job groups.
- **Made settlers actually walk the roads you build** — the route finder was blind to the road network, so a paved path only helped if someone already stood on it; a road tile now costs less to walk, so routes follow the network instead of cutting across open ground.

- **Fixed children holding adult jobs, and school with it** — the juvenile flag is set at birth and only cleared by graduation, so a child with a wrong flag stayed a grown-up: posted, armed, and refused as a pupil. Loading a save now returns them to childhood.
- **Stopped lending the energy glyph to fatigue** — the panel gave fatigue as a percentage where higher meant a worse penalty, beside a settler's energy percentage where lower means trouble, so it read as energy left; it now names the fatigue band and gives the penalty as a multiplier.
- **Fixed the family tree so it is about the person you opened it on** — the walk crossed every marriage into the whole colony and stopped at 200 names in visit order, so a settler's own child could go missing while a distant in-law stayed. Two generations either side now.
- **Added lineage grouping to the family owner** — settlers now group by who descends from whom through the parent links, so two unrelated settlers who share a surname no longer read as one family, and a line stays rooted in a founder who has died.
- **Fixed the family tree so it draws the whole family from any member** — one row per generation with parents, grandparents, siblings, spouses and descendants; the clicked settler is highlighted, the same family appears whoever was clicked, and cycles and remarriages no longer cut it short.
- **Put the child marker of the settler table back on the chip row** — a juvenile name no longer carries a bare emoji; a labelled child chip with a tooltip sits beside the no job and no home chips.
- **Fixed: "Civilization Builder" / "Species role" is gone from the citizen card — the block is deleted, not renamed again** — reported from play twice (*"why is she a civliazation builder then"* , then *"why stil civliazian builder"*). The first pass relabelled it: Role → Species role, moved under a Food chain · humans heading. That was a wording fix for a surface problem, so the second report was inevitable. The owner gave the decisive reason when confirming the deletion: *"work is already defined lower in the civ viewer and civilationd builder is nothign"* — the line was **redundant and empty at the same time** . Redundant, because this settler's work is already shown further down the same card as occupation plus the job and skill lines. Empty, because ecology.role is the **species'** trophic role, a constant (human: { role: 'Civilization Builder' }), printed on an **individual settler** , so it read as that person's job while saying nothing about them — every citizen, a newborn included, was a Civilization Builder. The whole block (Species role / Eats / Hunted, and the foodChainInfo table that existed only to feed it) is now **removed** . tsc clean, oxlint 0 warnings, and a repo-wide search confirms the string survives only in the comment recording its removal.
- **Fixed: the normal work day is 9 hours, not 8 — a colony working its own standard day was charged an hour of overtime every single day** — scheduleFatigue.NEUTRAL_WORK_HOURS was **8** while workSchedule.STANDARD_PRODUCTION_WORK_HOURS was **9** , so the game's own two halves disagreed about what a day is: a settler doing a standard 9-hour shift accrued FATIGUE_PER_EXCESS_HOUR (10) of fatigue for it and recovered nothing. getScheduleTargetHours caps its target at that constant, which is why the whole system shifted at once when it was corrected. The owner's rule is explicit (*"normal work day is 9 hrs"*), and the correction is self-evidencing: with NEUTRAL_WORK_HOURS = 9 the existing case *"emits one settlement summary for multiple meaningful work-fatigue increases"* — two settlers on a 12-hour shift — now produces **36** (30 excess − 4 base), which is exactly the value the test already asserted and which the 8-hour constant was silently defeating. Tests updated with the arithmetic recorded: a neutral nine-hour day recovers 30 − 4 = 26, one hour of overtime lands on 0 + 10 − 4 = 6, and a zero-hour day recovers 80 − (4 + 9×3) = 49.
- **Changed: a settler leaves early enough to be AT work when the shift starts, instead of starting to walk then** — owner: *"they should be AT work when working tim start not begining with walking"* . The allowance was a fixed WORK_COMMUTE_LEAD_HOURS = 1 for everyone, which cannot be met by a long leg, and COMMUTE_SNAP_DISTANCE only teleports commutes under 200 units — so anyone further out simply arrived late. simulation/humanMovement.commuteLeadHoursFor(distance, walkSpeed) now derives each settler's own lead from the movement owner's numbers (base speed, distRush, PATH_STEER_SPEED_RATIO), so it cannot drift from how the walk

behaves, and it is clamped to **1–6 hours** : the floor makes a short commute bit-identical to before, and the ceiling stops a settler being scheduled to leave the night before. **It is an estimate with a floor and a ceiling, not an arrival guarantee** — past MAX_COMMUTE_LEAD_HOURS a very long leg cannot be walked in the time available, and the existing shift-start snap is what still puts that settler at their post. Both facts are recorded in the function's doc because the first version of this conflated them.

- **Fixed: people get into and out of prison visibly — an imprisonment is its own event type with its own Chronicle filter** — reported from play as *"i cant seee if people get in prison its not happneing or its not logged and not shown"* . arrestForScandal logged 'event', so imprisonments landed in the generic Events bucket (1 531 of the owner's 2 000 Chronicle entries) with nothing to group them. There is now a real 'prison' type on GameEventLog, a **Imprisonments (jailed for scandal)** filter, a icon and colour, and — the part that made it invisible even when logged — a Jails count in the daily report, which previously showed neither scandals-adjacent number. **The tempting one-line fix ('event' → 'scandal') was deliberately not taken:** rumourLedger.SOURCE_KINDS reads type === 'scandal', so it would have pulled imprisonments into the rumour ledger and inflated the scandal count. The compiler found two more sites by itself — EventLogPanel's Record<GameEventLog['type'], string> maps for icons and colours errored immediately, naming the missing member. **A stale test was the reason this class of defect could hide:** tests/eventLogPanel.filters.test.ts hand-typed its list of event types and had already fallen behind (conception and divorce were absent), so the test guarding "every event type has a filter" could not have caught a missing one. It is now keyed as a Record<GameEventLog['type'], true>, which turns a new event type into a **compile error** there rather than a silent gap.
- *Verification for this entry: **npm run test:all green at** 245 files / 1493 passed, 2 skipped, 0 failed **, with** 0 lint warnings and 0 errors **over 350 files, and tsc clean on both the app and test projects. The two schedule panels and the new window were driven in a real browser (the repo has no DOM test tier): ordinary work accepted and committed 00:00–23:00, the Tavern accepted 04:00–23:00, the Hotel 06:00–21:00, the work-hours window opens and closes from its button, and the console was quiet throughout (0 console errors). The canonical browser bot still passes on the live game after the UI change (9 acts, farm before research, status line visible).** Not verified, stated plainly: **the walk-speed and TICKS_PER_HOUR rebase discussed in this session was** not **implemented — measured, the map is one settlement site (~102 m across at 11.76 px/m), a crossing currently costs** 23 ticks / 7.7 in-game hours **, and the owner's chosen target (one day to cross at TICKS_PER_HOUR = 5) requires walk speed** ≈ 14 px/tick with distRush deleted** , which rebases every duration in the game and is left as its own reviewed change. Footpaths, the night-hiding residual (~38 % still drawn), and the browser pass over the family-tree window and settler map rings also remain open.

UI, Gazette, Weather Visuals & Audio

- **Added:** a single persistent **Weather visuals** switch under Graphics, controlling all weather overlays, particles, fog, clouds, storms and seasonal particles without changing simulated weather.
- **Changed:** continuous integration now typechecks and runs the full unit test suite, in addition to source checks and the production build.
- **Fixed:** removed the Gazette hook dependency warning while keeping live Chronicle updates visible; newspaper summaries are only calculated when the Gazette is open.
- **Fixed:** restored the CPU profiler integration removed by a later update, repairing the build and restoring start, stop, live samples and the final worker report.
- **Fixed:** the CPU profiler docks above the map's lower-right controls instead of covering the build menu; expanding its report no longer moves the toggle.
- **Fixed: when the Chronicle reaches its retention limit, the oldest Gazette week is marked as potentially incomplete and its counts are shown as minimums.**

- **Fixed: loading a save no longer blacks out the game** — the toast list was dropped on load while the shell still expected one, so the first render after any load threw.
- **Changed: toasts are no longer written into a save** — a notification's life is measured on the real-world clock, so a restored one arrived already expired; the save keeps the dismissal ledger but not the toasts.
- **Fixed: the player's inspector no longer shows the dev-only simulation diagnostics, and stale copy names the current windows** — the drawer is gated on the debug flag, and the guide and tutorial no longer point at the retired tab rail.
- **Made a weather turn leave a record** — the weather changed silently, so the Chronicle never showed when conditions shifted; each change now logs a line.
- **Made the default test gate check types, lint and import cycles** — the default run ran neither a typecheck nor the cycle scan, which is how a committed type error and a runtime import cycle went unnoticed; it now runs both, and the long gate runs the year-long integration test.
- **Made the Guide and the shortcut sheet name the same six header doors** — each listed its own set, so the in-game help sent a stuck player to a door the header does not have; both now render one door list.
- **Made the work-hours editor and the family tree hold the keyboard** — the map's hotkeys stayed live behind those two windows and Escape acted on the game instead of closing them.
- **Vetted the weather on load** — an unknown weather or a non-numeric timer loaded into a valley whose weather could never change again.
- **Put the build hotkey on the collapsed build rail** — the only hint was a hover tooltip, which a touch player never sees.
- **Let the Guide open the shortcut sheet** — the sheet it pointed at could not be reached from the Guide, which has no hotkey of its own.
- **Made one despawn rule for finished caravans and departing visitors** — the carrier splice was the last manual removal beside the tick's own membership rebuild.
- **Made the volume preset reach the fallback music player, and bounded the master bus** — soft, normal and loud did nothing when the HTML audio fallback was playing, and the loud preset could clip.
- **Made the day rollover's entity rebuild linear** — the end-of-tick merge tested every new spawn against the whole living list, so the tick with the most arrivals did the most work; one membership set replaces the scan, keeping the same survivors in the same order.
- **Stopped a save choosing its own map size and calendar** — an untrusted tile dimension pair could ask for a 4000x3000 valley, and a forged ticks-per-day rescaled the whole timeline to the year 66666; both are now bounded to what this build actually writes.
- **Added Deno alongside npm** — the Deno 2.9 runtime is installed beside the existing setup, with deno:build and deno:test entries that drive the same single gate runner, so npm keeps the lockfile and nothing is replaced.
- **Made the Deno build produce a real desktop window** — it pointed at a browser-only React entry and died on a CSS import; it now packages the built game into a self-contained window that opens like the Tauri shell, with the unused dependency payload left out of the binary.
- **Recorded the owners the call graph showed missing** — the terrain subsystem, the Hunting Spot's pursuit and kill, the window shells and three single-purpose leaves had no ownership row, so 110 functions had no owner; the graph now resolves all 2767.
- **Fixed the year the player reads, so the colony's first year is Year 1 and not Year 0** — every notice, log line, header and panel now reads the year through one formatter that adds one; the stored year, its comparisons and the save format are untouched.

- **Put the test suite, the tooling tree and 59 bug reports under version control for the first time** — 316 files that existed only on this machine, taking the tracked count from 733 to 1112.
- **Added a clean command that empties the scratch directory wholesale** — 391 MB of scratch, 44 MB of build output and 162 MB of leaked browser profiles, with no keep-list, because an exception list turns scratch into storage.
- **Fixed (developer workflow): the dev server no longer dies on a locked scratch file, and an HMR parse error is what "fast-forward stopped working" actually was** — two findings from one vite crash log. **(1) The crash.** The server had been exiting (code 1) six times in a session, each time leaving a page that could not reload: chokidar threw EBUSY: resource busy or locked, watch 'C:\Wilderfolk\tmp\godot-ai-inspect\v4payload\addons\godot_ai\LICENSE' and the error took the whole process down. server.watch.ignored now skips tmp/ **alongside src-tauri/target/** — scratch is never imported by the app, so it has no business in the reload graph, and a locked file there can no longer kill the server. The **same fault had a second source** that killed it again straight after: the agent harness writes root files (that run: command.md) by atomically replacing them through a staging directory at the project root (.command.md.<pid>.<guid>.tmpdir/command.md.tmp), and vite watched that locked temp file and died — so editing the harness's own instructions took the dev server down. **/*.tmpdir/** and **/*.tmp are ignored too: nothing inside them is ever imported by the app, they are mid-write staging.** (2) The "broken" speed control. **The same log shows [PARSE_ERROR] Identifier 'venueGatherPosition' has already been declared in humanLeisureBehavior.ts at the moment a duplicate import existed during this session's editing — a failed transform breaks the module graph under HMR, so the page keeps its UI while the simulation stops advancing, which is exactly "the time fast forward doesnt work no anymre". It was fixed within the minute and the owner confirmed it cleared (*"nvm its solved"*); the measurement taken meanwhile is worth recording because it rules the speed path itself out: in a fresh build the** 10× button crossed a colony day in under 4 s **and the** 5 hotkey advanced 01:00 → 16:00 in 4 s** , i.e. about a day every 5 s from both the pointer and the keyboard — better than the 9–12 s/day baseline the handover recorded for 10×. tsc -p tsconfig.node.json exit 0 after the watcher change.
- **Fixed: one Escape press closed two windows — the keyboard claim is ordered now, and only the overlay on top answers Escape** — found by the browser probe on this change's first run, and it is the shape the new nested windows created: a subject window opens *from* the panel that lists it, so **two** overlays hold the keyboard at once, both bound Escape on window, and one press closed both. The probe reported it as "the second subject cannot be clicked" because the panel behind the first had gone with it. keyboardOwnership now records **claim order** (paint order here is interaction order: an overlay opened later is on top) and exports isTopKeyboardClaim(owner); useOverlayKeyboard leaves the key for the overlay above instead of acting on it. The overlay underneath keeps its claim — the game's own keys stay suppressed — and the next Escape closes it, which is what a player expects from a window opened over a panel. The probe's own assertion was the old bug's shape (it required *no* dialogs, which passed only because both closed) and is now the ported contract: the subject window is gone, the panel behind it is still open. tsc clean, oxlint 0/0.
- **Fixed: the dev server no longer full-reloads the page from Rust build output, so the game can actually be opened and driven in a browser** — vite.config.ts watched everything under the project root, including src-tauri/target/doc/**, which cargo doc fills with** thousands **of generated HTML files; Vite issued a full-page reload per file, so the page never settled. This is why the first browser run of the session could not reach the game at all. server.watch.ignored now excludes** /src-tauri/target/**. Nothing under target/ is ever imported by the app.

Engine Performance, Memory & Save Stability

- **Fixed:** the simulation worker is only enabled in supported runtimes, so main-thread fallback and diagnostics work without a browser Worker.
- **Fixed:** prison watch assignments keep the same weekly crew, including guards from overstaffed saves. Guards commute on weekends and off-screen; hourly post checks run after movement so arrivals count.

- **Added:** an opt-in in-game CPU profiler records call counts and CPU timing for simulation and rendering; export its report as JSON.
- **Fixed:** Stop now waits for the worker's final CPU sample before Copy, and the profiler control no longer covers the mini-map.
- **Added: a saved toggle to hide ordinary pop-up notifications without disabling the simulation, Chronicle or decision cards.**
- **Changed: human updates reuse workplace groups, social venue pools and entity lookups per tick, avoiding repeated scans and temporary coworker arrays.**
- **Fixed: every trade now advances when its workers turn up, not when the store accepts the goods** — a full store paid the crew no skill at all, so only farming produced masters while smiths, miners and teachers stayed at their entry skill; each staffed post now pays a shift of skill a day.
- **Fixed: a saved valley now regenerates as the valley it was saved from** — a compact save records the terrain noise seed it was generated with, and an older save with none falls back to the pre-change seed when its own completed fishing docks show the current one reclaimed their water.
- **Fixed: the drawn position no longer goes stale while a command is in flight** — the render cache was keyed on the tick alone, and a command's refresh arrives at an unchanged tick, so it served the previous frame's entities.
- **Fixed: a reloaded colony no longer re-adds an animation frame and a generation to every blade of grass and tree** — the load path defaulted both on every element, so one reload put them back and the next save paid for them again.
- **Fixed: a long pause no longer reads as a stalled simulation worker** — the silence clock kept ageing while the game was paused, so the first tick after a pause condemned a healthy worker and the game fell back to main-thread ticks.
- **Fixed: the opening cinematic and the valley choice behind it are back on every load** — the cinematic was always skippable in place, so suppressing it on a return visit only removed the map screen.
- **Changed: the mountain ground fill is now the re-exported PNG** — the bake and its boot preload name the file the art actually ships as, so the rock texture loads again.
- **Fixed: removed the 30-second auto-save — a manual save now always exports a JSON file, and the toggle now means save-on-exit.**
- **Fixed: the lifetime "Largest population" record no longer under-reports the peak** — it sampled only the year-end headcount, so a colony that peaked mid-year read a lower figure; it now tracks the peak every tick.
- **Added a lint and typecheck pass for the harness scripts** — the simulation probes and drivers sat outside every check, so their unused code and type errors went unseen; they are now reported by their own config.
- **Fixed the city benchmark so each tier holds its own population** — the per-tier target landed in the maintainer's minimum-headcount slot and was ignored, so the larger tiers ran down instead of staying steady.
- **Made a reload continue the same colony** — a save taken while the simulation ran in its worker recorded the main thread's random streams instead of the colony's, so a reload did not carry on the same deterministic sequence; the snapshot now travels with the saved world.
- **Made the voting-day gossip reachable** — it was gated to a tick remainder the ceremony can never land on, so the four "they are about to announce" lines could never fire; the roll now happens when the phase turns to tension.
- **Stopped a School and a Town Hall claiming their workers are assigned automatically** — the panel restated the manual-staffing rule by hand and omitted both, so a player waited for a teacher who could never arrive; it now reads the workforce owner's own verdict.

- **Stopped a malformed save's lists breaking the next day's food ledger** — the food history and the two dismissal lists had no rule at the load boundary, so a string in one of them was accepted and then threw inside a tick; each is now refused by name.
- **Stopped the work-hours panel and the taming list freezing at the values they opened with** — a projection memoised on the world object cannot see a tick when the simulation runs on the main thread; both now derive every render.
- **Made the venue editor quote the rule the tavern actually opens by** — the festival note was its own third copy of the override and could promise behaviour the simulation no longer had; it now renders the schedule owner's answer.
- **Checked every lifetime statistic a save carries** — only two of the fourteen fields were backfilled, so a damaged payload could leave the colony's counters undefined.
- **Made the Big News cards survive a save** — the cards were dropped while their dismissed-id list was kept, so a loaded colony's news panel came back empty.
- **Made a loaded colony's id bookkeeping consistent** — duplicate chronicle ids were kept beside live ones, and the allocator and entity-index repairs ran after the passes that resolve entities by id.
- **Stopped a load writing the player's auto-save preference** — a save's own setting is now adopted by the loaded colony only, never by the browser default the next settlement starts from.
- **Said which field a bad save file failed on** — the load-from-file path reported only "cannot be restored", and valid JSON that is not a save was blamed on a syntax error instead.
- **Made the save-slot check read the slot instead of parsing the save** — a returning player paid a full synchronous parse before the first frame.
- **Stopped the parked ecology ladder rewriting persisted state every day** — the disabled valley stage is now idempotent and normalises a legacy save once.
- **Repointed the architecture notes at the files that exist** — they still named a simulation file that was split into domain modules.
- **Loaded exactly the fill sprites the ground stamps** — the preload list named files nothing read and missed the ones the bake draws.
- **Refused a save whose container arrays cannot be read** — a malformed list among the world's arrays reached the interface and showed the player raw engine text instead of naming the field that was wrong.
- **Bounded the clock fields a loaded save may claim** — a negative, fractional or astronomically large tick was accepted, and the header, seasons and daily gate all derive a date from it, so a crafted file could freeze the calendar or name a year before the settlement.
- **Made a malformed venue window load as its documented default** — the tavern and hotel schedules were copied from the save unchecked, so unusable hours silently became the default and the bad value was written back by the next save.
- **Kept every critical alert on the strip and marked it without colour alone** — a flat two-chip limit hid the third of three simultaneous criticals behind "+N more".
- **Made a refused load leave the running colony alone** — the restore replaced the RNG seed part-way through, so a save that was then refused kept the failed load's randomness for the rest of the session; the session's seed and auto-save setting are now put back.
- **Refused save files with duplicate, fractional or negative ids** — two entities sharing an id made one of them vanish from every id lookup, unable to be selected or targeted, and the next save wrote the collision back.
- **Stopped a corrupted festival lasting forever** — the daily tick ends a festival when its remaining days reach zero, so a stored non-number made that test permanently false and the 1.5× festival boost never ended.

- **Told the truth when a save reaches a file but not the browser slot** — a full storage quota was swallowed, so the game reported success and then offered Load against a slot it had never written, which could restore an older colony instead.
- **Skipped the opening cinematic on a browser that has seen it** — it is a 23.6-second reveal that gated every load, so a returning player waited that long to reach Load; it plays once and stays reachable from the menu.
- **Made the high-speed report admit when it cannot see** — it averaged the simulation's clamped frame time, so a three-second frame printed as 100 ms; it now reports the raw frame maximum, ticks in flight, a pending command, banked lag, worker silence and page visibility.
- **Stopped a save reading the sim worker's host after a fault replaced it** — the export checked the host once, then awaited; a stall landing in that gap left the next line reading nothing, which threw while saving. The host is captured and re-checked by identity, as the load path already did.
- **Stopped one unappliable worker message from wedging the sim** — a tick or command result that arrived with no world attached was dropped without clearing its in-flight flag, so the game fell back to main-thread ticks and every save timed out waiting for the worker to settle.
- **The chronicle can now show its whole history on request** — the list drew the newest 500 rows with no way past them, which read as a 500-entry log; one button now loads the rest, and the rows are memoised so a full list stays cheap while the world ticks.
- **Fixed the chronicle export dropping its blob before the browser read it** — the download revoked its object URL in the same task as the click, so a large log could vanish with no error and no file; the revoke now waits.
- **Fixed a downloaded save's name disagreeing with the header clock** — a save taken in the colony's second year was named for the first, because the name printed the stored year while every other surface prints the year the player reads.
- **Fixed hot reload being disabled for the citizen card** — it exported a helper beside its components, which React Fast Refresh refuses, so every edit to it forced a full page reload.

□□ Bug Fixes & Codebase Refactoring

- **Fixed:** valley ecology transitions continue to enter the Chronicle when warnings are muted, without bypassing the cooldown for visible alerts.
- **Fixed: the worked-hours figure in the shift-fatigue line reads as the average it is** — "each worked 22.7h on average", not a shift length.
- **Fixed: two lifetime records that never recorded anything** — a colony's gathered resources and its most productive year stayed at zero for a whole game; gathering is now counted where storage accepts it, the best year is settled at each year boundary, and both show under Lifetime Records.
- **Fixed: a festival no longer pays innkeepers for the whole day** — the all-day tavern opening was credited as work, so every innkeeper worked a 24-hour day and sat at maximum fatigue, which cost the tavern a third of its output.
- **Fixed: a tamed animal that returns to the wild when its owner dies now leaves a Chronicle entry, so the change no longer happens silently.**
- **Changed: the stale one-off scripts under the archive folder are gone** — nothing referenced them and their imports pointed at modules that no longer exist, so the folder and the two exclusion clauses it needed were removed.
- **Fixed: a speech bubble's tail now reaches the speaker** — it was stopped three pixels short of its target even with nothing in its way, so it pointed at the head without touching it.

- **Changed: a hospital now takes five doctors instead of two** — the ward's crew cap rises so a full ward round and doctor-on-duty care can both be staffed at once.
- **Made auto-play fortify only a neighbour it has already paid** — the bot sends the food gift first and only raises a watchtower, barracks or wall when a hostile rival took that gift and stayed hostile.
- **Stopped auto-play raising a Bridge** — bridges are left to the player to place, so the bot no longer proposes one.
- **Made auto-play say what it is waiting for instead of calling an idle hour healthy** — the status line now names the larder, the beds, the study or the price behind the wait, in the game's own wording, and says the colony looks healthy only when nothing is short.
- **Made the ground re-bake at the zoom you are looking at** — a chunk now re-bakes in place when the zoom crosses a step, at most one chunk per frame, so a zoomed-out overview resolves at half resolution instead of sharpening every chunk at full detail; the enlarged maps stay smooth.
- **Made rivers reach the sea on the big maps** — a river could let out into a lake whose own channel climbs back over land, so the channel ended in a basin; an outlet now has to drain to a below-sea border cell, which holds at every map size and land type.
- **Let a research gift that fills the bar finish the project** — a visitor's gift clamped progress at the cap instead of completing, so the unlock arrived only on the next systems pulse; both gift paths now share the owner's completion rule.
- **Kept the map-setup gallery from clipping its own cards** — the second row of "Choose your land" was cut by the footer and pushed the size and difficulty choices off-screen at 900×600; the gallery scrolls inside a bounded height now, with the cue in its heading.
- **Gave the Prison four guard posts beside four cells** — the owner's cap: three nine-hour shifts still need three guards for round-the-clock cover, and the fourth takes rest days in turn.
- **Stopped the prison's own guards being jailed for scandal** — a caught affair jailed both offenders with no role check, so the sentence dissolved the very post the prison needs, and an unguarded prison leaks its prisoners.
- **Removed the dead founding-card state** — a deleted title card left a state slot and an unreachable branch; the live chapter card now has the name it actually renders.
- **Showed the biodiversity index against its own ceiling** — a bare number is now a score with an evenness bar a player can read.
- **Deleted the retired sidebar's styles** — the tab rail is gone, and its CSS was still shipped.
- **Removed a third Big News id allocator** — a private counter produced ids that the rollback contract and the dismissed-id list could not see.
- **Counted the colony where the daily tail uses it** — the tail reasoned about the population the day started with.
- **Rolled one prison escape chance per unguarded hour** — the daily pass stopped after the first escape, so a prison left open all day freed at most one prisoner.
- **Dropped four sprite paths that pointed at files which never existed** — those decor types draw procedurally, so the catalogue advertised images that could only 404.
- **Honoured the system's reduced-motion setting on the canvas** — pulses, sway, shimmer, sparkle and screen shake now hold still, while movement and floating text keep their timing.
- **Stopped selling upgrades that do nothing** — the Church, School, Tavern, Prison, Barracks, Hospital and Town Hall all offered an upgrade because they are staffed, but nothing reads their level; the rule now lists the types whose level really changes output or beds.
- **Counted a Town Hall official as working** — officials served their shift while the chronicle called every official day a rest day.

- **Fixed the mini-map's viewport box** — it was sized from the world instead of the canvas, so it was the wrong size at every zoom and ignored a window resize.
- **Stopped a notification self-destructing when you act on it** — clicking the card both jumped to the subject and dismissed it; the card now jumps and its own × dismisses.
- **Told the player how a prison sentence ends** — an escape and a release both only reached the map and the generic Events bucket of the chronicle; both now raise a notification and file under the prison filter, beside the jailing that started them.
- **Counted any guard on the post, not only the one the roster named** — a fully-staffed prison still freed prisoners, because an hour with other guards standing there read as unguarded and carried an escape roll.
- **Gave the prison four prisoner beds beside its three guard shifts** — its occupancy was the guard posts plus one cell, so a fourth guard left the jail nowhere to hold anyone, and two places counted the cells differently.
- **Called a parent's spouse a step-parent** — every married-in relative was named an in-law, which is why a toddler with a remarried father appeared to have two mothers.
- **Fixed rison escapes reading the roster instead of the post** — a guard asleep at home still covered his shift, so a full crew never leaked; every clock hour nobody stands on the post is now one escape chance, and a guard works his own shift's hours.
- **The Prison window shows the roster** — who holds the night, morning and afternoon watch this week, who is on post right now, and how many shifts are unstaffed.
- **The Prison runs on shifts** — three nine-hour watches (night, morning, afternoon and evening) with a one-hour handover cover the day, the crew rotates weekly on its own, and a prisoner can only slip out in an hour no watch covers.
- **A scandal now says so when it goes unpunished** — a caught affair with no guard on duty at the prison was refused in silence, so the whole jailing chain looked broken; it now writes one line explaining why, the first time it happens.
- **Archived the retired probes and the broken cycle tools out of the scripts folder** — they are parked with a note saying why, so the next reader extends the harness instead of writing another one-off probe.
- **Removed the empty right-hand sidebar** — the column under the inspector held one div and a comment and no controls at all, so it took width from the map for nothing; the inspector now has that height.
- **Put the family tree behind a small tree icon and widened its window** — the full-width row made the right column tall, so the label and the relative summary moved into the tooltip, and the window opens 44rem wide instead of the chat-sized 26rem.
- **Restored tabs in the work & venue hours window** — where three editors were stacked and scrolled, there is now an index you switch between, and every editor stays mounted so changing tab cannot discard a half-set hour window.
- **Added one test entry point with args, replacing seventeen npm scripts** — the gates are now one runner with a step menu, and the old names stay as aliases. Green: 245 files, 1495 tests, types and lint.
- **Fixed the lint gate, which named 70 files it never read** — the linter honours the ignore file, so listing the tooling directory linted nothing for a month; reading it surfaced 134 pre-existing findings, mostly missing node types.
- **Fixed the front page, which pointed at three files that do not exist** — and it described a tool removed months earlier. The developer commands left it; that page is for players.
- **Added a docs guard to the check step** — a link in a live document must resolve, and the newest changelog section stays short and names no files.
- **Added a ratchet so newly added code comments cannot carry history** — no dates, audit IDs or file references. The 173 existing audit citations stay untouched, so it never fires on old code.